Evaluation Technique of Cross Lingual Bengali Idiom Matching Using Multilingual Sentence Embeddings Without Translation

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under the guidance of

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19th June, 2026

OVERVIEW

Introduction

Objective

- Comparative evaluation of the outputs of the four sentence embedding models
- Prepare analytical visualisations and produce verdict based on evaluation scores

Models Used

- mSBERT, LaBSE, BanglaBERT, XLM-R

Core Principle

- No translation is performed at any stage; all comparisons use cosine similarity in a shared vector space

History

Key Research Papers that Inspired Our Work

Paper	Author/Year	What Inspired Our Work (Positive Side)	Limitations We Addressed In Our Work (Negative Side)
When Words Don't Mean What They Say; Figurative Understanding in Bengali Idioms	Adib Sakhawat et al. 2026	• Introduced a large Bengali Idiom resource with rich semantic annotations. • Highlighted the importance of figurative understanding and cultural grounding in NLP.	• Evaluation performed on **common standard models** rather than **BERT-based or Bangla-focused models.** • Focused on figurative meaning understanding within Bengali only; no **cross-lingual idiom matching.**
Alankaar: A Dataset for Figurativeness Understanding in Bangla	Geetanjali Rakshit, Jeffrey Flanigan 2025	• First Bangla Dataset focusing on figurative expressions with parallel Bangla-English translations. • Showed that figurative meanings are often not preserved in literal translation and require semantic representations.	• Focused on **metaphor identification** and **translation quality** rather than idiom-level matching. • Did not **evaluate multilingual sentence embeddings** for retrieval, similarity scoring, or clustering.
MAGPIE: A Large Corpus of Potentially Idiomatic Expressions	Hessel Haagsma, Johan Bos, Malvina Nissim 2020	• Created the largest English idiom corpus with a rigorous crowd sourced annotation pipeline. • Demonstrated the value of large-scale idiom resources for computational lingusitics.	• **English-only resource**, not covering low-resource languages like Bengali. • **Did not address multilingual semantic alignment** or cross-lingual idiom matching.

Proposed Architecture

Data Ingestion

JSON files are loaded into memory

Input Sources

- Bagdhara (Bengali Idioms)
- Bengali-Bangla (other rows)
- English Idioms
- Hindi Idioms

Alignment

Execute Strategy A (High Confidence) and Strategy B (Medium Confidence) to form:

- Ground-Truth Pairs (bn-en, bn-hi)
- Triplets (bn-en-hi)

Output

Aligned idiom pairs ready for embedding.

Embedding Generation

Use multilingual embedding models to convert idioms into semantic vectors.

Output

Vector representations for all idioms.

Evaluation Techniques

Semantic Similarity

Measure how semantically similar idiom pairs are using **cosine similarity.**

Cross-Lingual Retrieval

Retrieves the most relevant idioms across languages and evaluate **retrieval quality.**

Cross-Lingual Clustering

Cluster idioms in embedding space and evaluate **cohesion** of triplets.

Contribution of previous components

Inputs

- Sentence embeddings for bengali, hindi, english in (.npy) files
- Ground-truth pairs (bn-en, bn-hi) and triplets
- Similarity scores and collapse diagnostic
- A total of 23,544 unique idiom embeddings were generated across the three languages, representing the union of all idioms

Pair Type	Number of Pairs	Used In
Bengali-English (bn-en)	9,061 pairs	Similarity, Retrieval, Clustering
Bengali-Hindi (bn-hi)	4,096 pairs	Similarity, Retrieval, Clustering
Total idioms embedded	23,216 (bn+hi+en)	Clustering

Evaluation Tasks

Task	Purpose	Key Metrics
1. Similarity Scoring	Confidence in equivalent pairs	Mean cosine similarity, standard deviation, t-tests
2. Cross-Lingual Retrieval	Ranking ability	MRR, P@1, P@5, Hit@10
3.Cross-Lingual Clustering	Global space structure	Silhouette, Purity, Triplet Cohesion
4. Monolingual Bengali Clustering	Within-language quality	Silhouette (k=887)

Task 1 : Semantic Similarity Scoring

- Achieved by computing dot product of L2-normalised embeddings for every ground-truth pair.
- Uses mean cosine similarity across all pairs. A higher mean suggests stronger cross-lingual alignment.
- Reflects the model's overall confidence that same-meaning idioms are semantically close.

$$\mathrm{sim}(a, b) = \mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^{d} a_i b_i$$

Vector dot-product formula

Task 1 : Semantic Similarity Scoring

Model	bn-en mean	bn-hi mean	bn-en std
mSBERT	0.438	0.563	0.142
LaBSE	0.388	0.288	0.203
XLM-R	0.993	0.993	0.003 (collapse)

Table: Semantic Similarity Scores for Equivalent Idiom Pairs

Key Results

- mSBERT > LaBSE ($p<0.001$)
- bn-hi > bn-en only for mSBERT
- Reflects Indo-Aryan proximity

Task 2 : Cross-Lingual Retrieval

- Bengali query → rank all target idioms (English or Hindi) by cosine similarity
- Evaluation metrics penalise models that rank the correct match low. The metrics are : MRR, P@1, P@5, Hit@10.
- Sorts target idioms by descending similarity. Records the rank of the ground-truth match (if present).

Model	English rank (Build castles)	Hindi rank
mSBERT	12	5
LaBSE	8	11
XLM-R	1042	23

Sample ranking for idiom “আকাশ কুসুম”

Task 2 : Cross-Lingual Retrieval

Metric	mSBERT	LaBSE	XLM-R
MRR	0.0042	**0.0115**	0.0034
P@1	0.0016	**0.0065**	0.0009
Hit@10	0.0130	**0.0249**	0.0125

Table: Overall retrieval metrics

Key Results

- LaBSE wins retrieval.
- Dual-encoder + margin softmax excels at ranking.

Task 3 : Cross-Lingual Clustering

- Evaluates the global structure of the embedding space.
- All 23,544 idiom embeddings are clustered using K-Means algorithm (k=50).
- The following metrics are computed:
    - Silhouette Score (cluster separation, range -1 to 1)
    - Cluster Purity (language dominance – lower is better for mixing)
    - Triplet Cohesion (% of bn-en-hi triplets in same cluster)

Task 3 : Cross-Lingual Clustering

Key Results

- LaBSE → most language-agnostic (low purity, highest triplet cohesion)
- XLM-R → high purity but zero cohesion → pure collapse
- mSBERT → negative silhouette but non-zero cohesion

Model	Silhouette	Purity	Triplet Cohesion
mSBERT	-0.018	0.902	0.0106 (1.1%)
LaBSE	**0.041**	**0.745** (lowest)	**0.0176** (1.8%)
XLM-R	0.042	0.985	0.0000

Table: Clustering Results

Task 4 : Monolingual Bengali Clustering

- Evaluates the quality of the monolingual Bengali embedding space.
- Isolates the models' ability to organise Bengali idioms by their figurative meaning without any cross-lingual component.
- This task is particularly performed for BanglaBERT, which was pre-trained exclusively on Bengali.
- Bengali embeddings produced by each model is clustered with k=887 (approximately the number of distinct Bengali idioms divided by 10).
- Only Silhouette score is calculated.

Task 4 : Monolingual Bengali Clustering

Key Results

- LaBSE is best for fine-grained Bengali idiom discrimination.
- BanglaBERT has high mean similarity (0.831) but negative silhouette → compact but overlapping clusters

Model	Silhouette
mSBERT	-0.020
LaBSE	**+0.034** (only positive)
XLM-R	-0.051
BanglaBERT	-0.035

Table: Monolingual Clustering Results

Score Tally and Final Verdict

```
[1] CROSS-LINGUAL SEMANTIC SIMILARITY
  bn-en: mSBERT=0.4376  LaBSE=0.3878  XLM-R=0.9932  → XLM-R wins
  bn-hi: mSBERT=0.5629  LaBSE=0.2879  XLM-R=0.9934  → XLM-R wins
  [bn-en] Difference is statistically significant (t=7.9144, p=0.0)
  [bn-hi] Difference is statistically significant (t=146.6739, p=0.0)

[2] CROSS-LINGUAL RETRIEVAL
  MRR: mSBERT=0.0042  LaBSE=0.0115  XLM-R=0.0034  → LaBSE wins
  P@1: mSBERT=0.0016  LaBSE=0.0065  XLM-R=0.0009  → LaBSE wins
  P@5: mSBERT=0.0067  LaBSE=0.0180  XLM-R=0.0059  → LaBSE wins
  Hit@10: mSBERT=0.0130  LaBSE=0.0249  XLM-R=0.0125  → LaBSE wins

[3] CROSS-LINGUAL CLUSTERING
  silhouette: mSBERT=-0.0180  LaBSE=0.0408  XLM-R=0.0419  → XLM-R wins
  purity: mSBERT=0.9020  LaBSE=0.7453  XLM-R=0.9853  → XLM-R wins
  triplet_cohesion: mSBERT=0.0106  LaBSE=0.0176  XLM-R=0.0000  → LaBSE wins

[4] MONOLINGUAL BENGALI EVALUATION
  Self-similarity (mSBERT): 0.6814
  Self-similarity (LaBSE): 0.3307
  Self-similarity (XLM-R): 0.9942
  Self-similarity (BanglaBERT): 0.8306
    → Winner: XLM-R
  Clustering silhouette (mSBERT): -0.0200
  Clustering silhouette (LaBSE): 0.0338
  Clustering silhouette (XLM-R): -0.0509
  Clustering silhouette (BanglaBERT): -0.0350
    → Winner: LaBSE

=================================================================
  SCORE TALLY: mSBERT=0  LaBSE=6  XLM-R=5  BanglaBERT=0

  OVERALL WINNER: LaBSE
=================================================================

  CROSS-LINGUAL WINNER: LaBSE
```

Key Results

- Final Cross-Lingual Winner: **LaBSE**
- Different model might be suitable for specific tasks.
- **LaBSE** if the task requires a ranked list of possible equivalent idioms.
- **mSBERT** if the goal is to discover clusters of related figurative concepts across languages.
- **BanglaBERT** remains a strong choice for monolingual tasks that do not require fine-grained discrimination.

Score Tally and Final Verdict

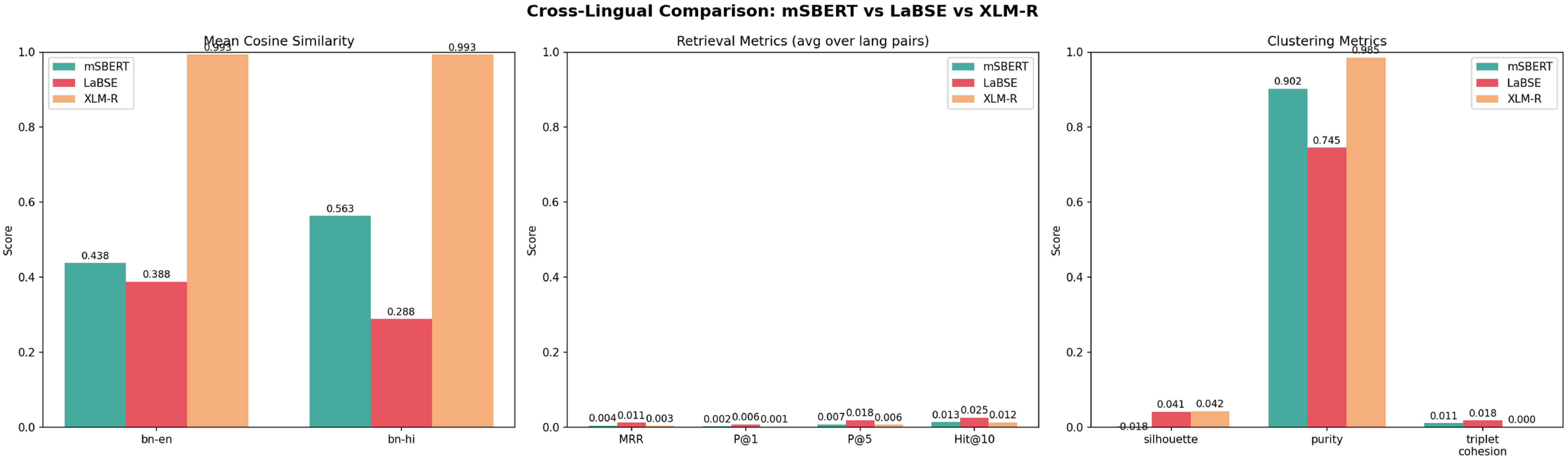

Fig: mSBERT vs LaBSE vs XLM-R - Full Model Comparison

Conclusion

- Both **mSBERT** and **LaBSE** capture cross-lingual idiom meaning without translation – significantly above **random baseline**.
- **LaBSE** excels at **retrieval** (ranking) and produces a more language-agnostic space (lower purity, higher triplet cohesion).
- **mSBERT** gives higher **similarity** confidence and correctly reflects Indo-Aryan proximity (bn-hi › bn-en).
- **XLM-R collapses** → caution for using general multilingual encoders without sentence-level objectives.
- **BanglaBERT** is strong for **monolingual** Bengali but cannot perform cross-lingual matching.

Drawbacks

- **Dataset imbalance** - skew affects clustering purity and retrieval difficulty.
- **Ground truth noise** - bn-hi alignment accuracy only 78% → up to 22% erroneous pairs.
- **No fine-tuning** - off-the-shelf models only → results are a lower bound.
- **Limited Hindi data** - restricts bn-hi evaluation breadth and statistical power.

Future Scope

- **Data enrichment** - expansion of hindi idioms, human verification of triplets.
- **Model adaptation** - fine-tune on verified idiom pairs, idiom-aware pretraining.
- **Methodological refinements** - trying different clustering algorithms, optimizing value of k.
- **Contextualised evaluation** - embed idioms in full sentences → more ecologically valid.

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Thank You